Type MS · Functions, procedures, parameters and libraries

Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 1 of 7 · find part c.ii in this mark scheme extract

5 (a)	A1 A2 A3 A4	2.1.4	Award 1 mark for each appropriate validation test (up to 2) and 1 mark for suitable example of erroneous data that matches the test e.g. First 3 characters are upper case letters: Fff123456o Characters 4 to 9 are non-zero numbers: FFF101111o Consists of only 10 characters: FFF111111ee Final character correct for sum of numbers: FFF112233o		(4)

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Functions, procedures, parameters and libraries

Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 2 of 7 · find part c.ii in this mark scheme extract

5 (b) (i)	B1 B2	1.1.9	Award 1 mark each for any of: • Simple implementation (1) • Can be used for sorted or unsorted lists (1) • If the target is at the beginning of the data structure the search will be faster than a binary search	Accept disadvantages of a binary compared with a linear search.	(2)

Type MS · Functions, procedures, parameters and libraries

Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 3 of 7 · find part c.ii in this mark scheme extract

5 (b) (ii)	B3 B4	1.1.9	Award 1 mark each for any of: • May need to compare with all items in list before search complete (1) • May need longer time for searching a large list (1)	Accept advantages of a binary compared with a linear search.	(2)

Type MS · Functions, procedures, parameters and libraries

Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 4 of 7 · find part c.ii in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Functions, procedures, parameters and libraries

Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 5 of 7 · find part c.ii in this mark scheme extract

5 (c) (i)	C1 C2	2.6.2	Award 1 mark each up to a maximum of 2 for: • Both can use arguments / parameter passing (1) • Both can use local variables (1) • Both can be called from anywhere within the program (1) • Code can be reused without being rewritten (1) • Both can be library files (1) • The code can be independently tested for both (1)		(2)

Type MS · Functions, procedures, parameters and libraries

Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 6 of 7 · find part c.ii in this mark scheme extract

5 (c) (ii)	C3 C4	2.6.2	Award two marks for a linked explanation such as: • A function must always return a result (1) whereas a procedure does not (1) • A function interface must have a data type (1) to signify the type of data that will be returned / a procedure does not need this (1) • The result of a function must always be used (either assigned to a variable or as part of a condition) (1) whereas a procedure does not explicitly return a result to be used (1) • A function produces information (1) whereas a procedure performs a task (1)		(2)

8 of 16

Type MS · Functions, procedures, parameters and libraries

Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 7 of 7 · find part c.ii in this mark scheme extract

Mark Scheme – Python Coding

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Functions, procedures, parameters and libraries

Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 1 of 7 · find part c.i in this mark scheme extract

5 (a)	A1 A2 A3 A4	2.1.4	Award 1 mark for each appropriate validation test (up to 2) and 1 mark for suitable example of erroneous data that matches the test e.g. First 3 characters are upper case letters: Fff123456o Characters 4 to 9 are non-zero numbers: FFF101111o Consists of only 10 characters: FFF111111ee Final character correct for sum of numbers: FFF112233o		(4)

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Functions, procedures, parameters and libraries

Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 2 of 7 · find part c.i in this mark scheme extract

5 (b) (i)	B1 B2	1.1.9	Award 1 mark each for any of: • Simple implementation (1) • Can be used for sorted or unsorted lists (1) • If the target is at the beginning of the data structure the search will be faster than a binary search	Accept disadvantages of a binary compared with a linear search.	(2)

Type MS · Functions, procedures, parameters and libraries

Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 3 of 7 · find part c.i in this mark scheme extract

5 (b) (ii)	B3 B4	1.1.9	Award 1 mark each for any of: • May need to compare with all items in list before search complete (1) • May need longer time for searching a large list (1)	Accept advantages of a binary compared with a linear search.	(2)

7 of 16

Type MS · Functions, procedures, parameters and libraries

Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 4 of 7 · find part c.i in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Functions, procedures, parameters and libraries

Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 5 of 7 · find part c.i in this mark scheme extract

5 (c) (i)	C1 C2	2.6.2	Award 1 mark each up to a maximum of 2 for: • Both can use arguments / parameter passing (1) • Both can use local variables (1) • Both can be called from anywhere within the program (1) • Code can be reused without being rewritten (1) • Both can be library files (1) • The code can be independently tested for both (1)		(2)

5 (c) (ii)	C3 C4	2.6.2	Award two marks for a linked explanation such as: • A function must always return a result (1) whereas a procedure does not (1) • A function interface must have a data type (1) to signify the type of data that will be returned / a procedure does not need this (1) • The result of a function must always be used (either assigned to a variable or as part of a condition) (1) whereas a procedure does not explicitly return a result to be used (1) • A function produces information (1) whereas a procedure performs a task (1)		(2)

Type MS · Functions, procedures, parameters and libraries

Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 7 of 7 · find part c.i in this mark scheme extract

Mark Scheme – Python Coding

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Functions, procedures, parameters and libraries

Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 1 of 8 · find part d.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (b)	B1	Award up to 2 marks for a linked explanation such as: • The number of keys are limited/only one shift used (1) making it easy to use brute force to decrypt (1) • It can be easy to find commonly used letters (e.g. E) (1) and guess the key (1)	Accept alternative similar wording.	(2)

Type MS · Functions, procedures, parameters and libraries

Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 2 of 8 · find part d.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (c)		Award 1 mark each up to a maximum of 4 for: <table><tr><td>Encrypted letter</td><td>f</td><td>l</td><td>m</td><td>k</td><td>t</td><td>r</td><td>w</td><td>h</td><td>e</td><td>e</td></tr><tr><td>Keyword letter</td><td>t</td><td>h</td><td>i</td><td>r</td><td>t</td><td>y</td><td>t</td><td>h</td><td>i</td><td>r</td></tr><tr><td>Decrypted letter</td><td>m</td><td>e</td><td>e</td><td>t</td><td>a</td><td>t</td><td>d</td><td>a</td><td>w</td><td>n</td></tr></table>		
	C1	Award 1 mark from: • flmktrwhee mapped to thirtythir (1) • Flmktr mapped to thirty (1)		
	C2	At least one letter decrypted correctly (1)		
	C3	At least one word decrypted correctly (1)		
	C4	Decrypted message 'meet at dawn' (1)		(4)

Type MS · Functions, procedures, parameters and libraries

Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 3 of 8 · find part d.iii in this mark scheme extract

Question	IMP	Answer	Additional Guidance	Mark
3 (d)(i)	D1	Award 1 mark for: • cipher/ letter / a single encrypted letter (1)	Do not accept word/message/text	(1)

Type MS · Functions, procedures, parameters and libraries

Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 4 of 8 · find part d.iii in this mark scheme extract

		• cipherLetter / a single encrypted letter (1)		(1)
3 (d)(ii)	D2	Award 1 mark for any of: • keywordLetter • plaintextLetter	Ignore case	(1)

Type MS · Functions, procedures, parameters and libraries

Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 5 of 8 · find part d.iii in this mark scheme extract

		• plaintextLetter		(1)
3 (d)(iii)	D2	Award 1 mark for any of: • subprogram that is already defined • subprogram that is already written • subprogram that is already compiled • subprogram that can be called without having to write code for it	Must be clear they are referring to subprogram provided by the language itself	(1)

Type MS · Functions, procedures, parameters and libraries

Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 6 of 8 · find part d.iii in this mark scheme extract

Type MS · Functions, procedures, parameters and libraries

Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 7 of 8 · find part d.iii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
3 (a)	A1	Get plaintext and store in plaintext variable	Accept alternative wording Line numbers may be different compared to the examples shown When testing the completed code use lowercase for the input	(1)
	A2	Get key and store in key variable		(1)
	A3	Validate key (must repeat until a valid key has been input)		(1)
	A4	Open file to write/append		(1)
	A5	Write cipher text		(1)
	A6	Close file		(1)
	A7	Displays ciphertext		(1)
	A8	Executing and producing correct output to file and screen (must have A3)		(1)
Code examples				
Python				

```
15
16 #       Add your code to get the plaintext and convert it to lowercase
17 plaintext = input("Enter the plaintext using lowercase letters: ")
18 plaintext = plaintext.lower()
```

```
# Add your code to get the key and make sure the key is between 1 and 25
while key < 1 or key > 25:
    key = int(input("Enter the key - a number between 1 and 25 "))
```

```
39 # Add your code to write the ciphertext to a text file
40 cipherFile = open("Cipher.txt", "w")
41 cipherFile.write(ciphertext)
42 cipherFile.close()
```

```
40
41 #       Add your code to display the ciphertext
42 print("The ciphertext message is:",ciphertext)
43
```

Type MS · Functions, procedures, parameters and libraries

Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 8 of 8 · find part d.iii in this mark scheme extract

Type MS · Functions, procedures, parameters and libraries

Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 1 of 11 · find part b in this mark scheme extract

Question		Answer	Additional Guidance	Mark
3 (a) (i)	A1	C# and Python monthNumber/monthsAndDays/count/found (1) Java monthNumber/monthsAndDays/count/found/input		(1)

Type MS · Functions, procedures, parameters and libraries

Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 2 of 11 · find part b in this mark scheme extract

3 (a) (ii)	A2	monthsAndDays (1)	(1)

Type MS · Functions, procedures, parameters and libraries

Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 3 of 11 · find part b in this mark scheme extract

3 (a) (iii)	A3	C# • WriteLine, ToString, ToInt32, ReadLine Java • Print, ParseInt, nextLine, ToString, Equals Python • print/len/int/input (1)	(1)

Type MS · Functions, procedures, parameters and libraries

Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 4 of 11 · find part b in this mark scheme extract

3 (a) (iv)	A4	Display/showMonthNameAndDays (1)	(1)

Type MS · Functions, procedures, parameters and libraries

Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 5 of 11 · find part b in this mark scheme extract

3 (a) (v)	A5	pMonth (1)	(1)

Type MS · Functions, procedures, parameters and libraries

Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 6 of 11 · find part b in this mark scheme extract

3 (a) (vi)	A6 A7	**Award 1 mark for each of:** **Type of test data** / **Test data** Boundary (1) / 1 Erroneous (1) / 15		(2)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Functions, procedures, parameters and libraries

Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 7 of 11 · find part b in this mark scheme extract

3 (b)	B1 B2	**Award up to 2 marks for a linked explanation such as:** • A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1) • (Using parameters) enables reusable solutions (1) using different data values/arguments (1) • Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1)		(2)

Type MS · Functions, procedures, parameters and libraries

Answer 4 of 6 · Question 3 part b · 2 marks

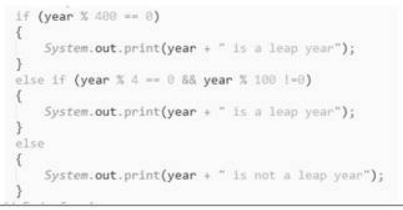
2021 NOV · Paper 02 · extract 8 of 11 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Functions, procedures, parameters and libraries

Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 9 of 11 · find part b in this mark scheme extract

3 (c)	**Award 1 mark for each of:**			
	C1	Condition to check whether the year exactly divides by 4 or equivalent condition (1)		
	C2	Condition to check whether year exactly divides by 400 or equivalent condition (1)		
	C3	Condition to check whether year does not exactly divide by 100 or equivalent condition (1)		
	C4	Correct message for at least one of the conditions (1)		
	C5	Program executes and produces the correct output (1)		
	C6	Efficient solution (1)		
				(6)
Code examples				
C#				
Java				

```
if (year % 400 == 0)
{
    Console.WriteLine(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    Console.WriteLine(year + " is a leap year");
}
else
{
    Console.WriteLine(year + " is not a leap year");
}
```

```
if (year % 400 == 0)
{
    System.out.print(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    System.out.print(year + " is a leap year");
}
else
{
    System.out.print(year + " is not a leap year");
}
```

Type MS · Functions, procedures, parameters and libraries

Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 10 of 11 · find part b in this mark scheme extract

Python	`if year % 400 == 0:`
	`  print(year,"is a leap year")`
	`elif year % 4 == 0 and year % 100 !=0:`
	`  print(year,"is a lear year")`
	`else:`
	`  print(year,"is not a leap year")`

Type MS · Functions, procedures, parameters and libraries

Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 11 of 11 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Functions, procedures, parameters and libraries

Answer 5 of 6 · Question 2 part a.i · 2 marks

2023 JUN · Paper 02 · extract 1 of 4 · find part a.i in this mark scheme extract

2(a)(i)	A1 A2	Award **one** mark for any of the following up to a maximum of **two** marks: • The code in the library has already been debugged/tested/optimised / library code less likely to have errors (1) • May include specialised/proprietary functions (1) • Library subprograms can be imported into code when needed / don't need to write code themselves (to save a programmer time/work) (1) • making it more readable / makes code easier to understand (Code files become shorter)	Allow characteristics of a specific library (math, random), if it can be attributed to unique bullets Don't allow quicker/easier/better without a reason being given Accept answers about subprograms as candidate may have written these themselves Line numbers are for guidance – mark answer as a whole	**2**
2(a)(ii)	A3	**The only correct answer is A** ***B*** *is not correct because iteration is not callable and does not return a result.* ***C*** *is not correct because a procedure, while callable, does not return a result.* ***D*** *is not correct because selection is not callable and does not return a result.*		**1**
2(a)(iii)	A4	Award **one** mark for: • A variable that exists/is accessible only in the subprogram/scope in which it is created (1)		**1**

Type MS · Functions, procedures, parameters and libraries

Answer 5 of 6 · Question 2 part a.i · 2 marks

2023 JUN · Paper 02 · extract 2 of 4 · find part a.i in this mark scheme extract

Type MS · Functions, procedures, parameters and libraries

Answer 5 of 6 · Question 2 part a.i · 2 marks

2023 JUN · Paper 02 · extract 3 of 4 · find part a.i in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(b)(i)</td><td>B1
B2
B3</td><td>Award one mark for each correct row.
<table><tr><th></th><th>Adult</th><th>Children</th></tr><tr><td>Normal</td><td>1/2/3/4</td><td>1/2/3/4/5/6</td></tr><tr><td>Boundary</td><td>1/4</td><td>1/6</td></tr><tr><td>Erroneous</td><td>Any value <1 or >4</td><td>Any value <1 or >6</td></tr></table></td><td>Accept all valid values for normal and boundary
Do not accept written numbers (two) as question asks for ‘numeric data’ – this is testing range, not data types.</td><td>3</td></tr>
<tr><td rowspan="5">2(b)(ii)</td><td colspan="2">Award one mark for each of:</td><td rowspan="5">Runtime error (B7) could be fixed before logic error (B6)
Applies to all languages
Accept < for <=
Allow /-1000 as alternative to B6
/1000 could happen after initial location e.g. print(“Order”, weightNeed/1000, “kilograms)</td><td rowspan="5">4</td></tr>
<tr><td>B4</td><td>Change =+= to +</td></tr>
<tr><td>B5</td><td>Change > to <=</td></tr>
<tr><td>B6</td><td>Change order of subtraction to: requiredWeight - inStock</td></tr>
<tr><td>B7</td><td>Change 0 to 1000</td></tr>
</table>

Type MS · Functions, procedures, parameters and libraries

Answer 5 of 6 · Question 2 part a.i · 2 marks

2023 JUN · Paper 02 · extract 4 of 4 · find part a.i in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Functions, procedures, parameters and libraries

Answer 6 of 6 · Question 2 part a.ii · 1 mark

2023 JUN · Paper 02 · extract 1 of 4 · find part a.ii in this mark scheme extract

2(a)(i)	A1 A2	Award **one** mark for any of the following up to a maximum of **two** marks: • The code in the library has already been debugged/tested/optimised / library code less likely to have errors (1) • May include specialised/proprietary functions (1) • Library subprograms can be imported into code when needed / don't need to write code themselves (to save a programmer time/work) (1) • making it more readable / makes code easier to understand (Code files become shorter)	Allow characteristics of a specific library (math, random), if it can be attributed to unique bullets Don't allow quicker/easier/better without a reason being given Accept answers about subprograms as candidate may have written these themselves Line numbers are for guidance – mark answer as a whole	**2**
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2(a)(iii)	A4	Award **one** mark for: • A variable that exists/is accessible only in the subprogram/scope in which it is created (1)		**1**

Type MS · Functions, procedures, parameters and libraries

Answer 6 of 6 · Question 2 part a.ii · 1 mark

2023 JUN · Paper 02 · extract 2 of 4 · find part a.ii in this mark scheme extract

Type MS · Functions, procedures, parameters and libraries

Answer 6 of 6 · Question 2 part a.ii · 1 mark

2023 JUN · Paper 02 · extract 3 of 4 · find part a.ii in this mark scheme extract

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Accept < for <=
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Type MS · Functions, procedures, parameters and libraries

Answer 6 of 6 · Question 2 part a.ii · 1 mark

2023 JUN · Paper 02 · extract 4 of 4 · find part a.ii in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark